

ASSEMBLYAI VOICE AGENT HACKATHON • SEP 2026

Readback

Say what you think you're signing.

A voice-verified signing protocol. It hears what you mean to sign, decodes what the transaction really does, and a Safe guard refuses anything that wasn't read back.

WHAT THE SCREEN SHOWED

A ROUTINE TRANSFER.

transfer • cold wallet → warm wallet

WHAT THE WALLETS SIGNED

A TAKEOVER.

operation 1 • DELEGATECALL → 0x9622...7242

Bybit, February 21, 2025. **\$1.46B**, one transaction.
Every signer checked it, on the channel the attacker
controlled.

THE FIX IS SEVENTY YEARS OLD

PILOTS READ EVERY INSTRUCTION BACK.

TOWER	“Climb to flight level three-five-zero.”
PILOT	“Climbing to three-five-zero.”
YOU	“Send one ETH to my cold wallet.”
READBACK	“Stop. This hands control of your Safe to an unverified contract.”

THREE REAL READBACKS • LIVE SITE • REAL ASSEMBLYAI VOICE

STOP “Claim my airdrop.” • A claim should give you something. This lets a wallet address with no history at all take your USDC, all of it, forever.

STOP “Send one ETH to my cold wallet.” • This doesn’t send anything. It hands control of your Safe to an unverified contract.

MATCH “Swap a hundred USDC for ETH.” • It will swap 100 USDC for at least 0.02 WETH through the Uniswap swap router, paid to you. Say confirm to sign.

Readback never shows you what a transaction does until you’ve said what you think it does. Otherwise you would just read it aloud.

01 · HEAR

YOU SAY IT.

AssemblyAI Voice Agent API

One WebSocket: speech, turn detection, tool calling, voice. The agent records your intent as a strict schema through a tool call. It never sees the transaction, so it can't be talked into agreeing with it.

02 · DECIDE

RULES CHECK IT.

Notary · @readback/core

Decodes the real calldata server-side, unpacks router multicalls and Safe MultiSend batches, checks every address on Base. 27 deterministic rules. Every finding names its rule. No model decides.

03 · ENFORCE

THE CHAIN REFUSES.

ReadbackGuard · Safe guard

A match on a Safe yields an EIP-712 attestation over the exact Safe tx hash. The guard refuses any transaction without one. Owner signatures alone are no longer enough.

RESEARCH FIRST: READ THE PRECEDENTS AT THE SOURCE

WHAT WE TOOK, AND WHAT WE REFUSED.

PRECEDENT	WE TAKE	WE REJECT
Safe Research Fiducia	Its cosigner transport: attestation appended to Safe signatures	Its hash zeroes gas fields; one cosign permanently allowlists (to, selector, op)
Safe Research Guardrail	Delegatecall allowlist, delayed guard exit	Rebuilding it
ERC-7730 clear signing	Breadth for protocol calls: +11.2% measured	Using it for attack primitives, where rules need typed semantics
AssemblyAI Voice Agent API	Tool calling as intent extraction	The voice as authority: enforcement never depends on it

BASE SEPOLIA • SAME BYBIT-SHAPED TRANSACTION • BOTH OWNERS SIGNED

WITH READBACKGUARD

REVERTED.

Implementation after: `0x29fc...C762`, Safe's own SafeL2 1.4.1. Untouched.

tx `0x8e3c182b...8bd8`

And a transfer attested by a spoken readback in the browser executed through the guard: `0x7ee42abc...0276`

WITHOUT IT

TAKEN OVER.

Implementation after: `0x60b7...c98A`, the attacker's contract.

tx `0x7919e8e9...e11`

15/15

spoken intents understood,
three voices, live API

15/15

verdicts spoken word for
word, checked on every
readback

24/24

guard tests on real Safe v1.3.0 and
v1.4.1, Base fork

1.9s

end of speech to
verdict, real browser.
Safe txs 2.9—3.4s

WHAT MEASURING CAUGHT

WE WERE WRONG, THEN WE MEASURED.

100%

false positives from our first “always block DELEGATECALL” rule on 81 real Base Safe txs. All went to Safe’s own MultiSend. Now unpacked: 77/77 real batches decode.

38.7%

of active Safes sent only transactions an open decoder could read. For the rest: “I can’t read this. Don’t sign something nobody can read.”

12ms

notary latency, down from 2,187 ms, by fetching chain facts while you’re still speaking.

READBACK

`readback-phi.vercel.app` · `github.com/Nuel-osas/readback`

Next: v0.2 relays audio through the notary so it hears the signer itself, closing the one trust gap v0.1 states plainly. Then an audited guard and a wallet SDK.